

Comprehensive Guide to Retrieval-Augmented Generation (RAG) — Page 1

1. What is RAG and Core Architecture

Retrieval-Augmented Generation (RAG) is an architectural pattern that enhances Large Language Models by dynamically retrieving relevant facts from private, external knowledge repositories before generating an answer. This prevents hallucinations, grounds answers in verified facts, and allows real-time private document queries without costly retraining.

2. Document Ingestion & Chunking Strategies

- Recursive Character Splitting: Breaks texts based on natural semantic boundaries (, , sentence terminators) with configurable overlap.
- Chunk Size: Typically 400 - 800 tokens. Smaller chunks preserve precision; larger chunks preserve context.
- Chunk Overlap: 10% - 20% overlap ensures semantic continuity across chunk boundaries.
- Metadata Enrichment: Every chunk must record document_name, document_id, page_number, chunk_id, and section headers.

Comprehensive Guide to Retrieval-Augmented Generation (RAG) — Page 2

3. Text Embeddings and Vector Databases

Embeddings are dense numerical vectors representing the semantic meaning of text in a high-dimensional space. Distance metrics such as Cosine Similarity measure semantic proximity.

4. Vector Database Architecture

- ChromaDB: Lightweight, open-source embedded vector database with persistent local indexing.
- Indexing Algorithms: Hierarchical Navigable Small World (HNSW) and Inverted File (IVF) index graphs.
- Metadata Filtering: Query filtering allows narrowing vector search to specific document subsets or timestamps.

Comprehensive Guide to Retrieval-Augmented Generation (RAG) — Page 3

5. Hybrid Search & Reciprocal Rank Fusion

Hybrid Search fuses dense vector semantic similarity with sparse lexical keyword matching (BM25). This solves semantic blindness to exact product codes, acronyms, and rare entity names.

6. Reciprocal Rank Fusion (RRF)

- RRF Formula: $RRF_Score(d) = \alpha * (1 / (60 + rank_vector)) + (1 - \alpha) * (1 / (60 + rank_bm25))$.
- Alpha Weighting: Alpha=0.5 provides balanced semantic and keyword precision.
- Reranking: Cross-encoders rescore top candidate passages for maximum relevance.